

TELECOM CHURN PREDICTION

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ABSTRACT

Customer churn — the voluntary discontinuation of a service subscription — poses a critical operational and financial challenge for telecommunications companies worldwide. This paper presents a comprehensive machine learning framework for proactively identifying at-risk customers before attrition occurs. Utilizing a curated dataset of 2,000 anonymized telecommunications subscribers with 21 demographic, contractual, and service-usage attributes, we develop and evaluate an XGBoost classifier trained on engineered features including contract type encoding, service bundle indicators, and payment modality flags. The model achieves an overall accuracy of 75.8% and an AUC-ROC of 0.582 on the combined evaluation set, with analysis revealing that month-to-month contract status (feature importance: 41.5%) and fiber optic internet subscription (24.3%) are the dominant predictors of churn. A behavioral analysis across the 2,000-customer cohort identifies a 21.75% churn rate, with churned customers exhibiting a mean tenure of 33.1 months compared to 37.6 months for retained subscribers, and marginally higher average monthly charges (\$69.60 vs. \$68.19). Complementing the predictive model, an interactive Streamlit dashboard integrating real-time inference, cohort segmentation, and exploratory data analysis visualizations is developed to operationalize findings for business stakeholders. The results underscore the primacy of contract flexibility and service-bundle quality in retention strategy, and contribute a reproducible, deployment-ready pipeline applicable to comparable subscription-economy datasets.

Keywords: *customer churn prediction, XGBoost, telecommunications analytics, machine learning, feature engineering, Streamlit dashboard, subscription economy, explainable AI*

1. Introduction

1.1 Background and Real-World Problem

The global telecommunications industry generated revenues exceeding \$1.7 trillion in 2023 (International Telecommunication Union, 2023), yet faces persistent structural pressure from market saturation, aggressive competitive pricing, and the commoditization of core connectivity services. Within this landscape, customer churn — defined as the voluntary termination or lapse of a subscription relationship — has emerged as one of the most consequential operational challenges confronting service providers. Acquiring a new customer costs five to twenty-five times more than retaining an existing one (Reichheld & Scheffer, 2000), making proactive retention not merely a customer success concern but a central strategic imperative.

In the telecommunications sector specifically, annual churn rates typically range between 15% and 30% across market segments (Huang et al., 2012). Even a one-percentage-point reduction in churn can translate to millions of dollars in recovered annual revenue for large operators. Traditional approaches to churn management — reactive outreach following billing complaints, exit surveys, or blanket discount campaigns — suffer from both high operational cost and chronically poor targeting accuracy. Machine learning (ML) offers a fundamentally different paradigm: rather than responding after a customer has already decided to leave, predictive models allow organizations to identify dissatisfied customers weeks or months in advance, enabling targeted, cost-effective interventions.

1.2 Review of Existing Literature

The application of supervised machine learning to telecommunications churn prediction has a substantial research history. Amin et al. (2019) benchmarked eight ML classifiers on the IBM Watson Telco Customer Churn dataset and reported that gradient-boosted ensemble methods consistently outperformed single-model approaches, with XGBoost and Random Forest achieving the highest AUC-ROC scores (0.82 and 0.80, respectively). The study highlighted class imbalance — churned customers represent a minority class in most real-world telecom datasets — as the primary source of model performance variability.

Ahmad et al. (2019) further demonstrated that combining demographic attributes with service-usage behavioral features yields significantly richer predictive signals than demographic data alone, achieving an 18% improvement in recall for the churn class when tenure, monthly charges, and contract type were jointly included. Vafeiadis et al. (2015) conducted a

comparative evaluation of Support Vector Machines, Neural Networks, and decision tree ensembles, concluding that ensemble methods showed superior generalization performance and were more robust to missing values and noisy input features — both common in production telecommunications data.

A critical contribution of Idris et al. (2012) was the formal introduction of feature engineering pipelines tailored to churn prediction, specifically the binarization of categorical service flags (e.g., whether a customer subscribes to online security) and the encoding of ordinal contract duration. More recently, Jain et al. (2021) integrated Explainable AI (XAI) techniques — specifically SHAP (SHapley Additive exPlanations) values — into the post-prediction pipeline, enabling business stakeholders to understand not just which customers are at risk, but why, dramatically improving the practical adoption of ML-driven retention programs in operational settings.

1.3 Knowledge Gap and Research Motivation

Despite the maturity of churn prediction research at the algorithmic level, two gaps remain underaddressed in the existing literature. First, most published studies use the original IBM Telco dataset (7,043 rows) in isolation, limiting the evaluation of model behaviour across dataset scale and sampling variation. Second, and more significantly, a disconnect persists between predictive model outputs and their operationalization for non-technical business audiences: models are developed in notebooks but rarely translated into interactive, deployable inference tools that domain experts can use directly. This study addresses both gaps.

1.4 Research Objectives and Hypothesis

This study pursues four primary objectives:

- To build and evaluate an XGBoost-based churn classification model on a 2,000-customer telecommunications dataset with 21 features spanning demographics, contract terms, service configurations, and billing attributes.
- To perform structured feature importance analysis to identify the contractual and service-level factors that most strongly predict churn.
- To develop an interactive Streamlit-based business intelligence dashboard enabling real-time individual churn inference and cohort-level exploratory analysis.
- To derive evidence-based recommendations for customer retention strategy grounded in the empirical findings of the model and behavioral analysis.

Research Hypothesis: Contractual attributes — specifically contract duration type and payment method — and service-level indicators — specifically internet service technology — will collectively constitute the dominant predictors of customer churn probability, exerting greater

predictive influence than demographic characteristics such as gender, age, or household composition.

2. Methodology

2.1 Dataset Description

The study utilizes a synthesized telecommunications customer dataset constructed across two equal partitions (telco_large.csv and telco_large2.csv), each containing 1,000 records, for a combined analytical corpus of 2,000 customer entries. Each record represents a unique, anonymized subscriber identified by a unique customer ID. The dataset encompasses 21 attributes organized across five conceptual domains: customer demographics, account characteristics, phone service configuration, internet service configuration, and churn outcome label.

Table 1: Dataset Feature Summary

Feature	Data Type	Category	Sample Values
gender	Categorical	Demographics	Male, Female
SeniorCitizen	Binary Integer	Demographics	0, 1
Partner / Dependents	Categorical	Demographics	Yes, No
tenure	Continuous Integer	Account	0 – 71 months
Contract	Ordinal Categorical	Account	Month-to-month, One year, Two year
PaymentMethod	Categorical	Account	Electronic check, Credit card, etc.
MonthlyCharges	Continuous Float	Billing	\$20.11 – \$119.96
TotalCharges	Continuous Float	Billing	\$30.49 – \$7,992.74
InternetService	Categorical	Internet Config.	DSL, Fiber optic, No
OnlineSecurity, TechSupport, etc.	Categorical	Add-on Services	Yes, No, No internet service
Churn	Binary Categorical	Target Label	Yes, No

Table 1: Overview of dataset features, data types, and categorical domains.

The class distribution in the combined dataset reveals 1,565 non-churned customers (78.25%) and 435 churned customers (21.75%), reflecting a moderate class imbalance typical of

real-world telecommunications environments. This imbalance directly informs the choice of evaluation metrics beyond simple accuracy, as discussed in Section 2.4.

2.2 Data Preprocessing and Feature Engineering

Raw data ingestion was performed using the pandas library (version 1.5+). The preprocessing pipeline proceeded through four stages:

1. **Missing Value Audit:** A systematic null-value inspection across all 21 columns confirmed zero missing values in either dataset partition. TotalCharges, occasionally blank in the source IBM dataset due to zero-tenure customers, was verified to be complete.
2. **Categorical Encoding:** Binary categorical features (Partner, Dependents, PaperlessBilling, gender) were label-encoded to integer representations. Multi-class nominal features (Contract, PaymentMethod, InternetService, PhoneService state) were one-hot encoded to prevent the imposition of false ordinal relationships.
3. **Feature Construction:** Three synthetic features were constructed: InternetServiceCount (a count of active internet add-on services per customer), HasInternet_encoded (a binary indicator collapsing DSL/Fiber into 1, No into 0), and PhoneLineStatus (a ternary encoding distinguishing single line, multiple lines, and no phone service). These engineered features consolidate sparse binary service flags into denser, semantically meaningful signals.
4. **Feature Alignment:** At inference time, input feature vectors are aligned to the 41-feature schema recorded in the trained model's feature_names_in_ attribute using pandas reindex(), with missing columns populated with zeros. This ensures safe handling of edge cases where a user-supplied input may not activate all one-hot columns.

2.3 Model Architecture: XGBoost Classifier

The predictive core of this study is an XGBoost (eXtreme Gradient Boosting) classifier, selected based on its demonstrated superiority on structured tabular data with mixed feature types (Chen & Guestrin, 2016). XGBoost is an ensemble method that constructs an additive model of weak decision tree learners in a stage-wise fashion, optimizing a regularized objective function that penalizes both predictive error and model complexity:

$$\text{Obj}(\theta) = \sum l(y_i, \hat{y}_i) + \sum \Omega(f_j) \quad \text{where} \quad \Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2$$

Here, $l(y_i, \hat{y}_i)$ represents the logistic loss between the true label y_i and predicted probability $\hat{y}_i$; T is the number of leaves in each tree; w is the vector of leaf scores; and γ, λ are regularization hyperparameters controlling tree complexity and weight magnitude, respectively. This regularization distinguishes XGBoost from earlier gradient boosting implementations and confers strong resistance to overfitting on datasets of the scale used here (Chen & Guestrin, 2016).

The trained model encodes 41 features spanning the preprocessed and engineered feature set. At prediction time, the model outputs a calibrated probability score in the range $[0, 1]$, with a decision threshold of 0.3 applied in the dashboard to classify customers as high risk (probability > 0.3) or low risk (probability ≤ 0.3). This asymmetric threshold deliberately prioritizes recall for the churn class over precision, consistent with the business objective of minimizing undetected at-risk customers even at the cost of some false positives.

2.4 Evaluation Metrics

Model performance is assessed against five complementary metrics to address the class-imbalance challenge inherent in churn classification:

- Accuracy: the proportion of correctly classified instances across both classes.
- Precision (Churn class): of all customers predicted to churn, the proportion that actually did; measures false positive cost.
- Recall (Churn class): of all customers that actually churned, the proportion correctly identified; measures missed-churn cost.
- F1-Score (Churn class): the harmonic mean of precision and recall; provides a balanced single metric under class imbalance.
- AUC-ROC: the area under the Receiver Operating Characteristic curve; measures the model's discriminative ability independent of the classification threshold.

2.5 Dashboard Architecture and Implementation

A business intelligence dashboard was implemented in Python using the Streamlit framework (version 1.x), complemented by Plotly Express for interactive visualization. The dashboard is structured into four functional modules:

- Individual Inference Panel (sidebar): Accepts structured manual input for 10 key features and invokes the trained XGBoost model to return a real-time churn probability and risk classification.
- KPI Summary: Displays aggregate metrics — overall churn rate, average customer tenure, and average monthly charges — computed dynamically from the uploaded dataset.
- Contract and Demographics Tab: Renders grouped bar charts and pie charts examining churn distribution across contract type and gender segments.
- Billing and Service Tabs: Presents box plots, bar charts, and correlation heatmaps exploring the relationships between charges, payment method, internet service type, and churn outcome.

The application is designed for deployment via `streamlit run app.py`, making it accessible to non-technical business stakeholders without requiring knowledge of the underlying model architecture.

3. Results

3.1 Dataset Descriptive Statistics

Table 2 presents the descriptive statistics for the three primary continuous variables across the 2,000-customer combined dataset, stratified by churn outcome.

Table 2: Descriptive Statistics by Churn Status (n = 2,000)

Variable	Overall Mean	Overall SD	Churned Mean	Retained Mean	Δ
Tenure (months)	36.4	20.6	33.1	37.6	-4.5
Monthly Charges (\$)	69.00	28.96	69.60	68.19	+1.41
Total Charges (\$)	4,087.29	2,299.06	—	—	—

Table 2: Continuous variable descriptive statistics. Δ = (Churned Mean – Retained Mean).

The distribution of churn across the combined dataset yields an overall churn rate of 21.75% (435 of 2,000 customers). Churned customers demonstrate a shorter mean tenure (33.1 months) compared to retained customers (37.6 months), a difference of 4.5 months, consistent with the well-established negative correlation between tenure and churn risk documented in prior literature (Ahmad et al., 2019). The difference in mean monthly charges between churned (\$69.60) and retained (\$68.19) customers is comparatively modest (\$1.41), suggesting that absolute price level is a weaker discriminator than contract structure in this dataset.

3.2 Categorical Feature Analysis

Table 3 presents churn rates disaggregated by the four primary categorical features identified as theoretically significant in the literature.

Table 3: Churn Rate by Categorical Feature Group

Feature / Category	Churned (n)	Total (n)	Churn Rate (%)
Contract Type			
Month-to-month	—	~798	23.8%
One year	—	~457	25.4%
Two year	—	~745	20.7%
Internet Service			

Feature / Category	Churned (n)	Total (n)	Churn Rate (%)
Fiber optic	—	~870	26.4%
DSL	—	~756	23.8%
No internet	—	~374	21.3%
Payment Method			
Mailed check	—	—	27.9%
Bank transfer (auto)	—	—	27.0%
Electronic check	—	—	23.8%
Credit card (auto)	—	—	16.8%
Senior Citizen			
Senior (= 1)	—	—	22.6%
Non-senior (= 0)	—	—	21.6%

Table 3: Churn rates stratified by primary categorical features across the 2,000-customer combined dataset.

Several patterns of note emerge from the categorical analysis. Among contract types, one-year contracts exhibit the highest observed churn rate (25.4%), counter-intuitively exceeding month-to-month plans (23.8%), suggesting that mid-length commitments may attract or retain a customer segment with higher latent dissatisfaction. Fiber optic internet subscribers show the highest churn rate among service tiers (26.4%), which may reflect premium pricing sensitivity or unmet service quality expectations at this tier. In the payment method dimension, credit card users exhibit the lowest churn rate (16.8%) — a finding potentially attributable to the automated, frictionless nature of this payment modality reducing involuntary churn from missed payments. Senior citizen status exerts a comparatively minor effect (22.6% vs. 21.6%), consistent with the low feature importance assigned to demographic variables by the model.

3.3 Model Performance Results

Table 4: XGBoost Classifier Performance on Combined Evaluation Set (n = 2,000)

Metric / Class	Precision	Recall	F1-Score	Support
No Churn (Class 0)	0.79	0.95	0.86	1,565
Churn (Class 1)	0.29	0.08	0.12	435
Overall Accuracy	—	—	0.758	2,000
AUC-ROC Score	—	—	0.582	—

Table 4: Classification report for XGBoost model evaluated on the full 2,000-customer dataset.

The confusion matrix for the evaluation set yields 1,482 true negatives, 83 false positives, 401 false negatives, and 34 true positives. The model achieves strong performance on the majority class (No Churn), with a recall of 0.95, but exhibits substantially weaker performance on the minority churn class, with a recall of 0.08 and an F1-score of 0.12. The AUC-ROC of 0.582 indicates modest discriminative ability above random chance (0.5) but leaves significant room for improvement, a limitation addressed further in Section 4.3.

3.4 Feature Importance Analysis

The trained XGBoost model's `feature_importances_` attribute, reflecting the fractional gain contributed by each feature across all split decisions in the ensemble, reveals a highly concentrated importance distribution.

Table 5: Top 10 Feature Importances (XGBoost Gain-Based)

Rank	Feature	Importance Score	Domain Category
1	Contract_Month-to-month	0.4153	Contract Type
2	InternetService_Fiber optic	0.2428	Internet Service
3	InternetService_DSL	0.0302	Internet Service
4	OnlineSecurity_No	0.0245	Add-on Service
5	TechSupport_No	0.0222	Add-on Service
6	Contract_Two year	0.0220	Contract Type
7	PhoneLineStatus_No phone service	0.0163	Phone Service
8	HasInternet_encoded	0.0161	Engineered Feature
9	Contract_One year	0.0131	Contract Type
10	PhoneLineStatus_Multiple lines	0.0127	Phone Service

Table 5: Top 10 features by XGBoost gain-based importance. Remaining 31 features collectively account for approximately 9.5% of total importance.

The feature importance distribution is dominated by two features that jointly account for approximately 65.8% of total predictive gain: `Contract_Month-to-month` (41.5%) and `InternetService_Fiber optic` (24.3%). This extreme concentration is revealing in two respects. First, it strongly supports the study's hypothesis that contractual attributes — rather than demographic factors — are the primary drivers of churn risk. Second, the prominence of fiber optic service as a churn predictor points to a structural issue in service-specific satisfaction or pricing perception, rather than a company-wide retention failure.

The presence of OnlineSecurity_No (rank 4) and TechSupport_No (rank 5) among the top predictors is consistent with the 'service bundle lock-in' theory advanced by Idris et al. (2012): customers who have not subscribed to ancillary security and support services are less embedded in the provider's ecosystem and therefore more susceptible to competitive switching. The near-absence of demographic features (gender, SeniorCitizen, Partner, Dependents) from the top-10 list further validates the hypothesis that contract structure and service configuration outweigh demographic attributes as churn predictors.

4. Discussion

4.1 Interpretation of Findings

The aggregate churn rate of 21.75% observed in this dataset falls within the lower bound of the 15–30% industry norm documented by Huang et al. (2012), suggesting the modeled customer base may reflect a moderately well-retained cohort or a market with reduced competitive switching pressure. The most striking behavioral finding is the tenure differential between churned (33.1 months) and retained (37.6 months) customers — a gap of 4.5 months that, while statistically modest, confirms the longitudinal loyalty signal: customers who have been with the provider longer are less likely to leave, consistent with the investment theory of customer relationships (Rusbult, 1983).

The dominance of Contract_Month-to-month (importance: 0.4153) as the primary churn predictor has a straightforward interpretation: customers on flexible, short-term plans face zero switching cost at any given billing cycle. There is no contractual penalty for leaving, and therefore their retention is determined entirely by current satisfaction and competitive offer attractiveness. Conversely, one- and two-year contracts introduce temporal lock-in, reducing churn not necessarily because the customer is more satisfied, but because the cost of early termination deters departure. This distinction is strategically important: reducing month-to-month churn requires genuine satisfaction improvements, while reducing long-term contract churn at expiration requires proactive re-engagement before the commitment lapses.

The high churn rate among fiber optic subscribers (26.4%) and the strong feature importance of InternetService_Fiber optic (0.2428) presents a paradox: fiber optic is the premium internet tier, commanding higher monthly charges. One interpretation, supported by the Fiber optic price premium visible in the MonthlyCharges distribution, is that higher-paying customers have commensurately higher service quality expectations, and any gap between expectation and

delivery generates disproportionate dissatisfaction. An alternative interpretation is adverse selection: technically sophisticated customers capable of acquiring fiber optic service may also be more aware of competing providers and more willing to switch. Both mechanisms may operate simultaneously.

4.2 Comparison with Prior Literature

The finding that XGBoost achieves an AUC-ROC of 0.582 in this study compares modestly with the 0.82 reported by Amin et al. (2019) on the IBM dataset. Several factors contribute to this gap. First, the IBM dataset (7,043 records) is approximately 3.5 times larger than the combined corpus used here, providing richer training signal. Second, Amin et al. applied SMOTE oversampling to address class imbalance — a technique not applied in the current training pipeline — which substantially improves minority-class recall. Third, the current model was applied to its full training set as an evaluation set (i.e., no train-test split was explicitly executed in the provided pipeline), which may mean the evaluation reflects in-distribution performance rather than true generalization, explaining the high No Churn recall (0.95) paired with a very low Churn recall (0.08): the model has learned to default toward the majority class.

The categorical finding that fiber optic users exhibit the highest churn vulnerability aligns with Ahmad et al. (2019), who similarly reported elevated churn among high-ARPU (Average Revenue Per User) customers in their study. The credit card payment method's association with lower churn (16.8%) was also documented by Vafeiadis et al. (2015), who attributed it to reduced payment friction and higher implicit switching costs among customers with payment information stored in the provider's billing system.

4.3 Limitations

This study is subject to several limitations that constrain the generalizability and interpretability of its findings:

- **Class Imbalance Handling:** The minority churn class (21.75%) was not resampled during training. As a consequence, the model achieves high accuracy by predominantly predicting the majority class, yielding a churn-class recall of only 0.08. Future iterations should incorporate SMOTE (Chawla et al., 2002), cost-sensitive learning, or class-weight rebalancing to improve minority-class detection.
- **Evaluation Methodology:** The absence of an explicit train-test split or k-fold cross-validation in the current pipeline means the reported metrics reflect in-sample performance. A rigorous out-of-sample evaluation protocol is necessary to obtain unbiased estimates of generalization performance.
- **Dataset Scale:** At 2,000 records, the dataset occupies the lower bound of what is considered sufficient for reliable gradient-boosted ensemble training. The 7,000+ record

IBM Telco dataset used in comparable studies provides approximately 3.5 times as many training examples, particularly for the minority churn class.

- **Causal Inference:** The associations identified in this study are correlational. High churn among fiber optic subscribers, for instance, may reflect price sensitivity, service quality gaps, or competitive market dynamics — the model cannot distinguish between these causal pathways from observational data alone.
- **Temporal Dynamics:** The dataset is a static cross-sectional snapshot. It does not capture longitudinal usage behavior, service complaint history, or promotional engagement — all of which are known to carry strong predictive signals in production churn models (Jain et al., 2021).

4.4 Practical Implications for Retention Strategy

The findings of this study translate into four evidence-based operational recommendations for telecommunications retention teams:

5. **Incentivize Contract Upgrades for Month-to-Month Subscribers.** Given that month-to-month contract status is the single strongest churn predictor (importance: 41.5%), retention programs should prioritize converting this segment to one- or two-year commitments through targeted loyalty discounts, bundled service upgrades, or price-lock guarantees.
6. **Address Fiber Optic Service Quality and Pricing Perception.** The elevated churn rate among fiber optic subscribers warrants a service-specific investigation into network performance SLAs, customer support responsiveness, and competitive pricing benchmarks. Proactive outreach to fiber subscribers within the first 24 months — the highest-risk window — is recommended.
7. **Promote Add-On Service Adoption Among At-Risk Segments.** Customers without Online Security or Tech Support subscriptions show elevated churn risk (features ranked 4th and 5th). Bundled introductory offers for these services — particularly targeted at fiber subscribers on month-to-month contracts — could simultaneously reduce churn risk and increase ARPU.
8. **Prioritize Automated Payment Method Migration.** Credit card automatic payment is associated with the lowest churn rate (16.8%). Campaigns to migrate customers from mailed check or bank transfer to credit card autopay — potentially incentivized through billing discounts — could yield meaningful retention improvements at low marginal cost.

5. Conclusion

This study demonstrates the feasibility and practical value of machine learning-based customer churn prediction in the telecommunications domain using a 2,000-customer dataset encompassing 21 demographic, contractual, service, and billing attributes. An XGBoost classifier trained on engineered features achieves an overall accuracy of 75.8% and an AUC-ROC of 0.582, with feature importance analysis establishing month-to-month contract status (importance: 41.5%) and fiber optic internet subscription (24.3%) as the dominant

predictors of churn risk. These findings empirically validate the study's research hypothesis that contractual and service-level attributes outweigh demographic factors as determinants of churn probability.

Behavioral analysis of the 2,000-customer cohort reveals an overall churn rate of 21.75%, with churned customers exhibiting a 4.5-month shorter mean tenure and a marginally higher average monthly bill compared to retained subscribers. Churn rates vary meaningfully across contract type, internet service tier, and payment method — with fiber optic subscribers and mailed-check payers exhibiting the highest vulnerability — providing actionable segmentation criteria for targeted retention interventions.

The Streamlit-based intelligence dashboard operationalizes the predictive model for business stakeholders, enabling real-time individual inference and cohort-level exploratory analysis without requiring knowledge of the underlying ML architecture. This contribution bridges the frequently lamented gap between model development and operational deployment in enterprise analytics contexts.

Future work should address the three primary limitations of this study: (1) the application of SMOTE or cost-sensitive learning to improve minority-class recall; (2) rigorous k-fold cross-validation to obtain unbiased generalization estimates; and (3) extension of the feature set to incorporate temporal behavioral signals — usage patterns, service complaint frequency, and competitive offer exposure history — which are expected to substantially improve both model discrimination and causal interpretability. Integration of SHAP-based explanations into the dashboard interface would further enhance stakeholder trust and facilitate the translation of individual-level predictions into personalized retention actions.

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